EXCEL ACADEMY
ELECTRICITY NOTES

⚡ Electric Current and Its Effects ⚡

1. The Basics: Electric Charge

Everything begins with charge! It's a fundamental property of matter, just like mass.

- **What is Charge (Q)?**
  - It's an intrinsic property of elementary particles (like electrons and protons) that causes them to experience a force when placed in an electric and magnetic field.
  - There are two types of charges:
    - **Positive Charge (+):** Carried by protons.
    - **Negative Charge (-):** Carried by electrons.
  - **Key Rule:** Like charges **repel** each other, and unlike charges **attract** each other.
- **Properties of Charge:**
  - **Quantization:** Charge is quantized, meaning it exists in discrete packets. The smallest unit of charge is the charge of a single electron, denoted by 'e'.
    - $e = 1.6 \times 10^{-19}$ Coulombs.
  - **Conservation:** Charge can neither be created nor destroyed; it can only be transferred from one body to another.
- **SI Unit:** The SI unit of charge is the **Coulomb (C)**.
  - One Coulomb is the charge contained in approximately 6.25×10^{18} electrons.

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2. Conductors and Insulators

Materials are classified based on how easily they allow charge to flow through them.

- **Conductors:** Materials that allow electric charge (electrons) to flow through them easily. They have a large number of free electrons.
  - **Examples:** Metals (Silver, Copper, Aluminum), Graphite, Salt solutions.
- **Insulators:** Materials that do not allow electric charge to flow through them easily. They have very few or no free electrons.
  - **Examples:** Rubber, Glass, Plastic, Wood, Pure water.

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3. Electric Potential and Potential Difference

For charges to flow, there must be a "push" or an "electric pressure." This is called potential difference.

- **Electric Potential (V):** The work done in moving a unit positive charge from infinity to a point in an electric field.
- **Potential Difference (V):** The work done to move a unit charge from one point to another. Think of it like a difference in water levels in two connected tanks; water flows from the higher level to the lower level. Similarly, charge flows between two points only if there is a potential difference.
  - **Formula:** Potential Difference $(V) = \frac{\text{Work Done (W)}}{\text{Charge (Q)}}$

$$V = \frac{W}{Q}$$

- **SI Unit:** The SI unit of potential difference is the **Volt (V)**.
  - **1 Volt** is defined as the potential difference between two points when 1 Joule of work is done to move a charge of 1 Coulomb from one point to another.
  - $1\ \text{V} = 1\ \text{J/C}$

- **Voltmeter:** An instrument used to measure potential difference. It is always connected in **parallel** across the points where the potential difference is to be measured. It has a very **high resistance**.

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4. Electric Current

The flow of charge is what we call electric current.

- **What is Electric Current (I)?** The rate of flow of electric charge through a conductor.
  - **Formula:** Current (I) = $\frac{\text{Charge (Q)}}{\text{Time (t)}}$

$$I = \frac{Q}{t}$$

- **SI Unit:** The SI unit of electric current is the **Ampere (A)**.
  - **1 Ampere** is defined as the flow of 1 Coulomb of charge per second.
  - $1\ \text{A} = 1\ \text{C/s}$
- **Direction of Current:**
  - **Conventional Current:** The direction of flow of **positive charge**. This is from the positive terminal to the negative terminal of a battery in the external circuit.
  - **Electronic Current:** The actual direction of flow of **electrons**, which is from the negative terminal to the positive terminal.
  - **By convention, we always use the direction of conventional current in circuit diagrams.**
- **Ammeter:** An instrument used to measure electric current. It is always connected in **series** in the circuit. It has a very **low resistance**.

5. Ohm's Law

This is one of the most fundamental laws in electricity, relating potential difference, current, and resistance.

- **Statement:** At a constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends.
  - $V \propto I$
  - This gives us the famous equation:

$$V = IR$$

where **R** is the constant of proportionality called **Resistance**.

- **Resistance (R):** The property of a conductor to resist the flow of charges through it. It is the ratio of potential difference across the ends of a conductor to the current flowing through it.
  - $R = \frac{V}{I}$
- **SI Unit:** The SI unit of resistance is the **Ohm (Ω)**.
  - **1 Ohm** is the resistance of a conductor through which a current of 1 Ampere flows when a potential difference of 1 Volt is applied across its ends.
- **V-I Graph:** The graph between potential difference (V) and current (I) for an ohmic conductor is a straight line passing through the origin. The slope of the V-I graph gives the resistance (R).

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6. Factors Affecting Resistance

The resistance of a conductor depends on four main factors:

1. **Length of the Conductor (L):** Resistance is directly proportional to the length. Longer wire = more resistance.
   - $R \propto L$
2. **Area of Cross-section (A):** Resistance is inversely proportional to the area of cross-section. Thicker wire = less resistance.
   - $R \propto \frac{1}{A}$
3. **Nature of the Material (ρ):** Different materials have different resistances. This property is called **resistivity** or specific resistance.
4. **Temperature:** For most conductors, resistance increases with an increase in temperature.

Combining these factors, we get:

$$R = \rho \frac{L}{A}$$

- **Resistivity (ρ):** The resistance of a conductor of unit length and unit cross-sectional area. Its SI unit is **Ohm-meter ($\Omega \cdot m$)**.
  - Resistivity is a characteristic property of a material; it doesn't depend on the dimensions of the conductor.
  - Metals and alloys have very low resistivity. Insulators have very high resistivity.

Ohm's Law: The Fundamental Relationship

German physicist Georg Simon Ohm established a fundamental relationship between voltage, current, and resistance for many materials. **Ohm's Law** states that, under constant physical conditions (like temperature), the current flowing through a conductor is directly proportional to the potential difference across its ends.

Mathematically, this is expressed as:

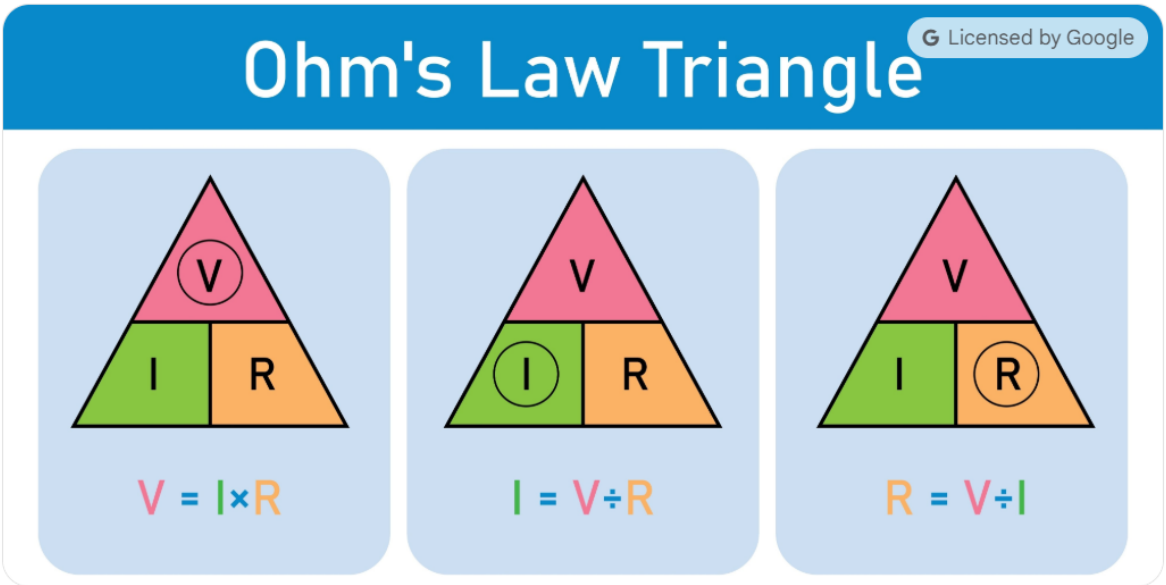
$$V \propto I$$

or

$$V = IR$$

where:

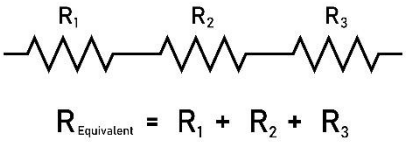
- V is the potential difference in volts.
- I is the current in amperes.
- R is the resistance in ohms.



This simple yet powerful equation is the cornerstone of circuit analysis.

4. Combination of Resistors

Resistors can be combined in two main ways in a circuit:



Resistors in Series

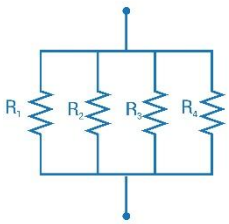
When two or more resistors are connected end-to-end, they are said to l

- **Current:** The current is the **same** through each resistor.
- **Voltage:** The total potential difference across the combination is the **sum** of the potential differences across individual resistors. (V=V1+V2+V3)
- **Equivalent Resistance (Rs):** The total resistance of the circuit increases.

$$R_s = R_1 + R_2 + R_3 + \dots$$

Resistors in Parallel

When two or more resistors are connected between the same two points, they are said to be i



- **Voltage:** The potential difference is the **same** across each resistor.
- **Current:** The total current is the **sum** of the currents flowing through individual resisto
- **Equivalent Resistance (Rp):** The total resistance of the circuit decreases.

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Key Takeaway: We connect devices in **parallel** in our homes. This ensures that each appliance gets the full mains voltage (e.g., 220V) and can be operated independently with its own switch.

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5. Heating Effect of Electric Current

When current flows through a resistor, some of the electrical energy is converted into heat energy. This is known as the **heating effect of current** or **Joule's Heating**.

Joule's Law of Heating

It states that the heat ('H') produced in a resistor is:

1. Directly proportional to the **square of the current** (I2).
2. Directly proportional to the **resistance** ('R').
3. Directly proportional to the **time** ('t') for which the current flows.

- **Formula:**

$$H = I^2Rt$$

Other forms of the formula are: $H = VIt$ and $H = \frac{V^2}{R}t$.

Practical Applications

- **Electric Heater, Iron, Toaster:** These work on this principle. They use a heating element made of a high-resistance alloy like **nichrome**.
- **Electric Bulb:** The filament is made of **tungsten**, a metal with a very high melting point. It gets white-hot and emits light.
- **Electric Fuse:** A safety device consisting of a wire with a low melting point. If the current exceeds a safe limit, the fuse wire melts and breaks the circuit, preventing damage to appliances.

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6. Electric Power

Electric Power (P) is the rate at which electrical energy is consumed or dissipated in an electric circuit.

- **Formulas:**

$$P = VI$$

$$P = I^2R$$

$$P = \frac{V^2}{R}$$

SI Unit: The SI unit of power is the **Watt (W)**.

  - **1 Watt** is the power consumed by a device that carries 1 A of current when operated at a potential difference of 1 V.

Commercial Unit of Electrical Energy

The commercial unit of energy is the **kilowatt-hour (kWh)**, also known as one 'unit'.

- **Definition:** 1 kWh is the amount of energy consumed when an electrical appliance of 1 kilowatt power rating is used for 1 hour.
- **Relation to Joules:**

1 kWh=1000 W×3600 s

1 kWh=3.6×106 J

Calculating Electricity Bill: Cost = Total kWh consumed × Rate per kWh

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7. Household Circuits (ICSE Focus)

Electricity from the mains is supplied to our homes through three wires.

The Three Wires

1. **Live Wire (or Phase):** This wire is at a high potential (e.g., 220 V). It is usually **brown** in color.
2. **Neutral Wire:** This wire is at or near zero potential. It completes the circuit. It is usually **blue** in color.
3. **Earth Wire:** This is a safety wire connected to a metal plate buried deep in the earth. It is usually **green** or **yellow** in color. It provides a path for current to flow to the earth in case of a fault, preventing electric shock.

Safety Devices

- **Fuse:** As explained earlier, it's a safety device with a thin wire that melts and breaks the circuit during **overloading** (drawing too much current) or a **short circuit** (when live and neutral wires touch). A fuse is always connected to the **live wire**.
- **Miniature Circuit Breaker (MCB):** This is a switch that automatically turns off when the current in a circuit exceeds a safe limit. It can be reset manually, unlike a fuse which needs to be replaced.

Wiring System

Appliances are connected in **parallel** in a house. The **ring system** of wiring is often used where the live, neutral, and earth wires form a complete loop or ring from the distribution box around the rooms and back. This allows multiple sockets to be connected without using very thick wires.